# ActInf GuestStream #021.1 ~ Adam Safron

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hello everyone it's april 25th 2022 we're here with adam saffron and today we're going to be having a guest stream the radically embodied conscious cybernetic bayesian brain from free energy to free will and back again so adam thanks a ton for joining on this day really looking forward to your presentation and thanks again adam take it away uh thank you for inviting me and um or letting me come to talk with you and see i'll present some ideas from a uh recent manuscript in entropy called the radically embodied conscious cybernetic bayesian brain from free will from free energy to free will and back again and after that looking forward to discussing some of the ideas with all of you and um hopefully this is the beginning of an ongoing conversation so what i tried to do with this manuscript was basically um marry good old-fashioned cognitive science with a more an activist perspective trying to find a middle way between these ways of viewing of things and this is the view that i've come to over a number of years where um the idea being that embodiment like at this point you know it's not news that the mind is embodied but uh the claim is that it is uh very very embodied some would say radically embodied um i'm not sure if that made the most sense to use that term because the term radical with respect to embodiment is usually pointing to a perspective or radical in activism that avoids talking about representations or models and instead of saying you know what's really going on is these couplings of the world and maybe you can say there's like an implicit information processing or intelligence even that might be objected to but we talk of representations and symbols and models from cognitive science um that's those are not real things and so this paper goes through a perspective in which basically um we do uh have representations and models and we also have um these sort of inactive couplings with the world um and i say it's radical and that uh nearly uh every mental process you would care to talk about the way you have to understand it according to this view would be first and fundamentally and foremost in terms of its embodied character so before i move on as everything coming across clearly uh just okay yeah i did disable my video because due to some incredible consequences of different situations i'm using a hotspot but it's all good you're on the run let's review a little background i mean a lot of this um uh people who were watching this with but i'll review some uh background on predictive processing and uh active inference um maybe some of what i'll present is a somewhat heterodox view that's also something to discuss so a basic account of predictive coding you have these deep pyramidal neurons and you have these superficial pyramidal neurons where they're deep with respect to cortex as having this layered structure roughly six layers with some variations on the theme depending on where you look and the idea is that the cortex as a whole is engaging in what's sometimes called hierarchical or what's often called hierarchical predictive coding or potentially predictive processing because there might be some assumptions with saying predictive coding that not everyone would sign up for such as that you are only suppressing and explaining away information as opposed to sometimes enhancing it and there's other nuances that could make people object to saying predictive coding instead of getting into those weeds uh the idea would be that you have this hierarchy of beliefs implicit and neuronal organization um the nature of these beliefs and this gets into these debates about representation and inactivism but this idea that you can say there's an implicit belief at each neuron and set of neurons about what they expect what patterns they expect to encounter and according to this predictive processing view through their activity they're trying to predict this incoming activity and suppress it or explain it away to minimize overall activity and you that you only update these beliefs where you got it wrong via a prediction error and this is a under certain certain assumptions if you have a good model and events in the world are relatively slow relative to your internal signaling units you can have substantial energetic savings for an organ of about two percent of your body mass and 20 of your metabolism seems like the kind of thing natural selection would do and there's some decent evidence for this this so the basics of neural architecture here for cortex is viewing it as a predictive processing hierarchy or a header and that each of your modalities would be a different belief hierarchy and they would all come together and exchange their information for a joint belief across all your modalities in terms of what you think um should be in your sensorium as it moves through the world uh whether it's the world you're in right now or some sort of counterfactual world in the present or future so that's predictive processing in a nutshell and so in this account the goal was to use predictive processing to try to get into the micro mechanics of goal-oriented behavior and agency and so what i suggested is that um the work has been done within active on this sort of imaginative plan pursuing of goals the work to refer to would be sophisticated inference or sophisticated affect of inference which is thinking about basically planning into the future as this tree search of possibilities where you're going on different branches of these possibilities and then picking the ones either in imagination or in reality that would correspond to you minimizing your prediction error the best or realizing value the best but this exploration of a tree of possibilities and going down different branches and so this would be i suppose a particular flavor of this and the idea being that um a lot of what would be pulling along this stream of consciousness of imaginings um would be this iterative comparison between sensed and imagined states or between various goal states so here um i have this comic book that a friend helped me make and i have multiple comics throughout i tend to view comics as being really highly expressive in a way that um i think we should probably use them more in terms of being able to capture the the joints and experience space with sufficient richness and detail and there could even be something like comic book-esque about the stream of experience which could be part of the reason that they're so effective but to return to this comic or a modal frame sequence as i sometimes call it um you see in the bottom here this is the actual pose of the agent and up top what's going through the mind's eye whether we're talking about an imagination or a perception and so here it's just a simple contrasting of what you're perceiving in each given moment and then what you're imagining so you're sitting you want tea then look and you say i'm not drinking tea i'm at my computer and there's a laptop in front of me you look down and you're imagining having tea in your cup but then you look and no there is indeed no t and then you imagine a next step and so what's what's happening here is you're doing this contrast and what i propose is that in contrast between these states they're arranging they're basically setting up prediction errors for each other where what doesn't match would potentially get you a natural prioritization of what features are unaligned and then from those you use that to retrieve some sort of relevant affordance to help you bridge this gap between your imagination reality and there's a lot of devils in details of the ways this could work but that some sort of iterative contrasting goal states in a goal hierarchy where it's like you might have an ultimate goal but to achieve that goal you have to now go and think of what are all of the sub steps and the sub goals that would be needed to actually get the tea in my hand you're going to have to ambulate to the kitchen make the tea etc etc and as you're going back and forth at some place some point you either have a plan that's compelling enough or the next action is close enough to your present pose that you just find yourself launching into action sometimes accompanied by a feeling of having decided sometimes you just find yourself moving but that's the basic um account here that the stream of consciousness is being pulled along in this goal-oriented fashion this value-driven fashion um sometimes more freely wandering but sometimes but but usually there's an interest and you're as an active inference and predictive processing your preferences or your attracting states the tractors that you'll find as a system which you're attracted to takes the form of predictions and so the things in your mind that you're imagining that they become the goal state they become the state of the world that you prefer in that instance or in general if it's a regular kind of imagining and this now becomes the attracting state that your system self organizes and you engage in action selection of different kinds to minimize that difference between what you're imagining and what you are estimating for the world um i go into some discussion of uh neural architecture and different aspects that might be more or less relevant for different um of this kind of action selection um i probably should have just you know thrown this on top of a picture of a brain um but i didn't but in some ways it might be a good thing and that i wonder whether this account of having a dual-tier system something like uh a system in a system two is a broader story than just brains and so you know i have written out here like here would be neural substrates of what would enable you to do this but um well it could be across brains you might have similar processes happening within let's say like an insect nervous system by different means like instead of the basal ganglia you might be using the mushroom bodies um if we're talking like an ant colony for instance it's possible you have a different like attracting states of the colony where some are taking this role like a higher level controller and a lower level controller but the basic idea here is that you have an upper level controller that's capable of aggregating information across different so you have this lower level process by in this control hierarchy where the lower level is taking care of your immediate couplings with the world this would be um the more inactive portion um if we're talking about the brain it would be something like um loosely assembled um constellation of cortical reactive dispositions may be orchestrated by the cerebellar system of what would be called amortized inference and it would be largely unconscious and automatic um phenomenologists would talk about just like um things being ready at hand as opposed to being present to hand you're just effortless engagement and effortless mastery with but on top of this um you would have a slower process in this case i'm saying it's more involving conscious processing something more system two-like to use kahneman's language um where it's involved in more deliberate potentially effortful uh modes of cognition which would be a higher level of control over this um faster smooth coupling system go into some details of neural architecture potentially this might be just what you need for a lot of active inferential agents something like this across whether or not we're talking about neurons this this might be a broader story for um active inferential agents so to return to the example um of making tea uh you go into it somewhere saying that um you know areas associated with mental imagery maybe even physical and computational substrates of mental imagery phenomenal phenomenality itself but like the petunias for instance um involved in imagination have heavily implicated um multiple reasons to think why this would be the case but you know i have this little cartesian theater here with the different imaginings down below the idea is you are iterating through different imaginations largely via large-scale orchestration by the hippocampal complex and it's a capacities to couple with the rest of cortex to guide these high-level attracting states where some portions of which would be your experience and so the reason you would bring up the hippocampus in here is it seems to be a specialized subsystem for sequential predictive modeling you could say that for cortex as a whole technically hippocampus is a kind of cortex arcade cortex old vintage but that basically um so demacia will talk about this in terms of a convergence divergence zone where many things will funnel in and then funnel back with high centrality um uh i forget his name he said waterloo um the spawn architecture um but uh he'll talk about semantic pointers but there's different ways of viewing the this the high centrality of the campus as it basically will predict this trajectory of the overall organism through space you can both encode memories what was happening in my thorium whether in reality or imagination at each point as i was moving moving through space and time and you could also use it for replay including in counter factual circumstances and so um i have some diagrams here at the neural systems involved and right here you would have the i think that's good for now so um everyone wants to know more details about that later we could to return to um trying to go more phenomenological um so these modal frame sequences are comics ultimately what i'm trying to build up is a sufficiently detailed account of neural architecture and a sufficiently detailed account of phenomenology that we can talk about so that we can basically talk about something as highfalutin as free will and have some purchase so in this a little more complex modal form sequence you have that up top an imagining you know row below that you have what's actually in your perception below that you have your pose your proprioceptive imprints and then below that you have your enteroceptive inference as you're stepping through the stream of consciousness is evolving through space and time encoded or orchestrated by the hippocampal system at this high potentially most integrative high level of action selection are having different wounds in your body corresponding to the enteroceptive competence of um or the interceptive aspects of um what it means to be in different situations so for instance like a hunger situation here you're getting prediction air intercepted prediction air that would be um roughly vaguely localized here's like a cool sensation um around your um gut and mouth or thirst or hunger something like this and now as you're going into movement you might have here like um you're still having this enteroceptive code of hot and cold states pleasant and unpleasant but it's kind of getting infused into other other modalities where your proprioception some of the phenomenology of like feeling like an urge in your body it's one of the claims i make is that you have this kind of quasi-synesthetic cross-mapping between enteroceptive signals signals and the proprioceptive and exterceptive asp um their associates uh part of the reason i claim this and say that this is both part of the phenomenology of will and maybe part of the mechanism also um is that interceptive signals um are poorly localizable in space and time they tend to have pretty low resolution for them and so they may drift free from their sources and as a kind of anomalous inference but adaptively adaptive anomalous inference you they may get infused into the relevant um effector and sensor systems and objects in the worlds for which um they would be relevant you know i make a lot of different claims in this work uh from a number of years of study a lot of it can be taken a la carte like you can find some things to be more compelling some things be less compelling i do think you get a whole lot greater than some of its parts but it can also be taken in a piecemeal fashion so you might not be particularly compelled by account of interception as being synesthetic in nature um but nothing really not too much will ride on that but it could will write on that but it could end up being important so i'm putting it in there and so across all of this is happening within a perspective of computational neurophenomenology as a neil seth calls it or i call it marion neurophenomenology and the reason i call it marion is specifically his account of multiple levels of analysis which i guess super vein on each other and you can do two things because things have these multiple levels that they can be engaged with um you can either engage with them a la carte because of uh let's say for instance you want to do cognitive science but you have no idea how the brain works and then for many years we haven't and there's a lot of mystery still you can kind of bracket like here's this big set of unknowns but it's at a different level but working at this other level we can obtain knowledge even though we might be ignorant at these other levels um and that would be you know functionalism and cognitive science but uh for me what's more exciting about this actually this marian stack is the capacity not for independence but interdependence and mutual constraints using the implementational level or the mechanistic level as a constraint of what's a likely set of algorithms using a likely set of algorithms to test hypotheses on the implementation level using what you think the functions things like neuropsychiatric literature to constrain and then your imaging literature to constrain algorithm and thereby implementational so they the idea of what excites me is like building in this crossroad puzzle constraints and then cross-referencing them within the phenomenon level i'm using basically machine learning as a person principle and typically uh going to the races of this idea that a lot of cortex could be understood in terms of historical predictive problems uh but also evoking some additional machine learning principles but that's the overall uh scope of it in terms of the um like all the body parts and faces this is part of like the radical embodiment idea is that instead of like the cortex being this like grab bag of associations with areas and does that mean um the idea is that once you really place it front and center as a cybernetic controller for an embodied embedded agent once you place embodiment front and center wait a second that's not too many steps removed from where the ears first pipe in so the idea that something like you know going from phonemes or you know going from sound patterns undifferentiated to phonemes it ends up making cortex and its functions in my opinion a lot more applicable it becomes somewhat like location location location and location relative to the modalities that ends up explaining a lot of things a similar story for broca's area um what a coincidence that it's not too many you know it's a hop skip and a jump away from what might uh move your mouth bits there's doubles in details here right i don't be glib and underestimated complexity but the claim is that actually once you place embodiment in the center an elegance kind of falls out and becomes a lot easier to keep track of things if you're thinking of cortex as this control cybernetic control hierarchy over to basically help a body move through the world and model the world as it moves through it so we are ready i think to get to free will um and so taking these modal frame sequences and here is doing one for this experience of standing um to make tea and then taking that model and then pouring it to a bad experience so what is a little bit paradigm um so some people say le bay some people say look back i think that's deadlocked but i'll continue to say well bacon is more common and i just will um but the idea is that you're there and you want to see do people or don't they have something like meaningful conscious control over their actions and so the idea is okay let's strip this down and take like the most insignificant of all actions like well surely we need to have control there um if we're gonna say we have control anymore and so the decision to just move your hand based on feeling like you want to do so um and this coming from basically it's endogenous you're not being it's not being decided from without but um you doing something based on something internal to you and what was found with these um lebet's studies or the bay is that you get this ramping activity um preceding the decision to move your hand so it's one more thing you have this clock right and there is it's a big clock that you're looking at and there might be a dot or a big hand moving in a circle and the reason that you have this is basically so that you can estimate the time when it was that you said i will move my hand when did you decide and you then say okay when you made that decision what were you looking at where was this dot as it moved around in a circle and this can get you very fine resolution um sub second resolution because if it's a big clock that's because then the dot will be moving slowly and you can say okay it was right but what's observed is that you're getting this ramping activity with this predictive of the action called the readiness potential and it will peak you know a good 200 milliseconds before you reporting it varies but it'll peak substantially before you saying i've decided and so if and so the claim is well if this potential has predictive information of your action and is the cause of your action and it's happening prior to your conscious decision your decision cannot be the thing that caused the action there's a there's a problem right away with this in that with erp methods you're actually averaging over many many trials to get these waveforms because you're measuring from the outside of the brain it's like trying to detect like events inside a stadium while measuring like listening to the roar of the crowd on the outside and you're limited what you can do so you have to average together a lot of different um episodes and then you get these event-related potentials or erps and but so you get predictive information from this of when people moved but the idea that a readiness potential is causal or deterministic in some fashion there's some debate about that whether you can infer that i'm going to say the readiness potential is in fact reflecting a causal process and it's reflecting a conscious causal process even that is contributing to the action um upstream of um the decision itself where the decision could have some more you know it depends i don't know if there's one account but it could have some epithelial aspects to it and that the decision is not necessarily um the thing is it's not the only uh difference maker there but the idea is that this is a function of mental imagery where you are imagining and as you're rehearsing moving your hand you're having different things associated with it and that building up of that recurrent a building up of activity in these body maps whether we're talking about the getting ready to actually change your proprioceptive pose as you're imagining starting to move closer to a threshold where you're going to get robust enough activity that is actually going to drive your muscles and your effector systems or building up of activity in your anteroceptive modalities like your insular and cingulate hierarchies and maybe their cross maybe their synesthetic like cross couplings with other modalities that that building up of a feeling of urge is itself consciousness leading to the stream the causal stream of action that i would say so yeah the and then on top of all the different imaginings would be orchestrated by the capital system would be the basic model currently and so and and so currently i am working on uh expanding this um to seeing what different states of consciousness um the ways in which they could impact these sorts of phenomena um in terms of changing your awareness of this rehearsal process uh or changing uh that'll be one of the primary uh things of interest and also whether differences in the mechanisms of agency could potentially contribute to the phenomenology of psychedelics i'll try to explain what i mean um so here would be someone relatively unaltered engaging in mental rehearsal their minds started wandering off to nature they're seeing a tree here's imagination here's perception here more vivid than imagination and they're just sort of imagining things imagining will i move won't i move and then at a certain point imagination becomes vivid enough that they can really see it happening and at a certain point of uh vividness of imaginings as a function of this rehearsal um of will i move won't i move you get enough build up of whether we're on the implementation level we might talk or mechanistic level we might talk about recurrent activity in terms of active inference on the functional level we might talk about model evidence but at a certain point you cross a certain threshold leading to model of you altering your proprioceptive pose and this becoming a self-fulfilling prophecy through active inference you then minimize the prediction error between your imagination i am moving and the and the action through reflex arcs the way that um basically motor control is thought or is thought to work in active inference and predictive processing so the psychedelic angle would be as someone is dosing with these compounds their imagination becomes more vivid and they might show stronger ramping at first and maybe greater awareness of this patterns of mental stimulation leading up to the action it might get stronger still as you dose more and more also greater coupling between anteroceptive and proprioceptive modalities increasing the gain on this positive feedback between mental imagery influence of urge and so in a way this would be an enhancement of willing the willing aspect of agency as you move up this dose response curve but then you get to a certain level and you start to get into this psychedelic regime where your thought becomes highly creative and then your directed willing might start to go down um potentially the idea is you might find other things competing for your awareness different creative uh possibilities so you're now imagining nature very vividly and this is competing with you rehearsing will i move or won't i move [Music] you directing extremely high doses you might get and compromise of these mechanisms of agency of this conjoining of your enteroceptive and your proprioceptive predictions of your basically of your stream of consciousness being pulled along by value-driven policy selection via sophisticated effect of inference so i think so the reverse direction is this um in terms of uh this paradigm potentially informing psychedelics could be that one source of hallucinations or delusions or um atypical um altered uh consciousness and cognition would be that if you take the basic mechanic micro mechanics of agency and and you compromise that so you know as you're you're beginning to somewhat imagine an action but it's not well integrated into some causal controlling you could come to this anomalous inferences about what's causing these basically um the different um endogenous activity your brain the different like uh events that could lead to motor to motor actions if these aren't being integrated into a self-consistent coherent causal unfolding one of the ways that you could try to make sense of this would be a confabulation of something like a hallucination and so this could also explain some of psychedelic phenomenology um and potentially um a clinical neuropsychiatric phenology with things like schizophrenia as potentially being a function of disrupted agency mechanisms so that's currently the area of focus and there's going to be a pre-print just came out on that called on the degrees of freedom worth having psychedelics is a window for understanding and expanding uh free will i believe that's what i called it a little bit um also apologies if there's any strange stream hiccups but could you maybe unpack what the alien or extraterrestrial was doing and how does the confabulation relate to a narrative understanding of action so in this one you know we're starting out just like imagining your hand moving or imagining being in nature and then same here as the dose goes up and it goes up and you might start to get some more intense enteroceptive sensation um sensations uh the five ht2a receptors which um heavily mediate the effects of classic psychedelics uh they're they're highly expressed in the interior insulin the top of the anteroceptive hierarchy and so you keep going up becomes the sensations become more intense imagination becomes closer to reality and so here like what i'm calling the heroic dose in this extreme dose regimes you're starting to get for instance like your ownership of the actions are starting to get undermined so here i'm depicting like you find your hand moving but you're not accounting for your willing of the move being the thing you're not saying oh i was rehearsing it and it felt this way and that was part of the cause of streaming into action you just found yourself moving and so one way you could make sense of this or confabulate around this here is a hand like it's a feeling of the hand pushing down on your hand from without like just an example of a potential confabulation or hallucination someone might have and this one here it's expanded and it's this basically someone it's not even your hand anymore pressing it but an alien's pressing it it's just an example of for instance um so psychedelics they the classic psychedelics will act in these five ht2a receptors they are heavily expressed on these deep pyramidal neurons um of quarter texas six layers uh the layer five um well basically they're big neurons that loop with the thalamus and they let you form these broad synchronous complexes which in predictive processing are thought to encode your beliefs some subset of which would be your experience and so if you increase the gain on these uh the idea is that you might have enhanced perceptual vividness but because though you've done a radical alteration to of multiple kinds to like the normal regime of processing in your mind you're the normal ways you would model even though you're experiencing things intensely this would uh potentially be of an atypical variety of experience and so here it's the intense experience the intense perception at perception of the alien it's a counter factual alien it's not actually an alien uh but that would that would be an example of an anomalous inference you might make and it has some um relation to what's happening and the fact that maybe there was some motor rehearsal going on but you're not able to keep track of it and so and here this is um you're actually having phenomenal consciousness in each of these moments but you don't have conscious access unless in these iterative moments of experience you can somehow get call backs where um things can be contextualized within uh some sort of causal unfolding the claim would be that basically this would be a way of separating out phenomenal consciousness from conscious access that there's things that can happen as you're generating these quail states as estimates of what's happening of system world system world system world specifically your sensorium as you think it is based on your current engagement or in counter factual circumstances and that this that's the stream of consciousness but that not all the frames of experience are accessible or rememberable and so in this case here you know maybe there was some rehearsal but you're not keeping track of it and then the the um the the best causal account you were able to come up with was that an alien made you do it and you might have even forgotten that this was you or maybe you do remember it but would need probably some more contextualization or not it's unclear but yeah so that would be the basic idea is that psychedelics would be increasing consciousness level or the vividness of but potentially undermining the coherence in terms of causal unfoldings or your usual priors by which you structure the world um in the spirit of um robin carrera harris and carl firstens rebus model which i'm currently discussing in the context of a broader framework that i call albus but yeah i don't know did that help at all or like like yeah it also it reminded me of what you said about the uh different levels and their interdependence and mutual constraints so it's like a sound is heard and potentially in one state of consciousness or mind or brain it's 98 likely to be one thing and option b and c are one percent likelihood and so just goes forward with as if option a like as if i'm in control but then some things might happen where like that um one percent chance that it is an alien gets accepted and sort of passed forward and then that might lead to the experience through the generative model of something that like was a distributional so it's like a change in quantitative distributions that leads to a qualitative change at increasingly higher and categorical and semantic experiential layers beautifully said i thought and it would depend on your um cultural priors so and whether we would call this like a relaxed belief or a strength and belief i'm not sure uh but you know for instance you might not be seeing in like if you you know or an indigenous people of south america for instance um you know depend they might not see an alien it might be um related to something having to do with their spirituality and their their whatever available myths you have would then go into basically this interpretation your best guess which is always wrong um sometimes useful but your best guess of what's going on will be different depending on um you know what they would call the the set in the setting but also where um the setting is the broader cultural of the sentence setting are also broader to the culture not everyone will be not everyone will come to the inference of aliens probably and actually that's that's been shown um that that hallucinations are uh culturally specific yeah with which kinds of extraterrestrial morphologies are seen in relationship to which are prevalent in the media and the stories of a given region i don't know all the details but one can imagine yeah like not everyone i mean like you could like say like um maybe there is something about like gray aliens which is like a little more archetypal i don't know if i'd say this but like uh if you look at like the eigen dimensions like morphological variation what's relevant to us like you're just like exaggerating many of the dimensions that would be relevant for us engaging in inter-subjective modeling or modeling of ourselves even for the sake of control it's like hands like bigger hands might be important um where is like an attentional inference like where is this person attending to where am i attending to so you don't want to know the heads and the eyes that's going to be really crucial um yeah that's that's a far out story you know there could be aspects of uh that aren't just through the medium but it's it's super interesting and um uh yeah not not to go into alien stream mode but um thinking about a large headed archetype with very large dark eyes and the long fingers like you're kind of hinting at it's like it's all pupil it's all cortex it doesn't have the strength but it has a lot of nimbleness with its hands and with its body and so like that's like one archetype of of cognitive action it i don't know and then what maybe there's a blob and a shadow there's like these different archetypes different little characters in this comic book of cognition not that different from a child's affordances too right like they're not they're not capable of like moving things around much but it's like their intentional actions are mostly like hand-eye coordination of like that's their purchase on the world that's their grip but you know for all this you know so this is the current like focus is how might psychedelics inform our understanding of the micro mechanics um and how might the micromechanics of agency inform our understanding of what psychedelics are and that's currently uh one of the things i'm focusing on and about to do some experiments on um at this center for psychedelic consciousness science i'm working out at hopkins but the fundamentally upstream of this uh the idea is that these studies which purportedly um show that there's no free will uh they show no such thing uh it's very ambiguous what they're showing and i believe if you do an active inferential handling of them uh you'll actually see that there are several ways in which consciousness is contributing to causal streams leading to action uh which would be having some of the varieties of free will worth having um you know is it arbitrary degrees of free will no uh are there some varieties of free will worth having that we don't have like can you just like fly all of a sudden that'd be nice it'll be worthwhile uh we can't do that either but if you want your beliefs and desires to be able to meaningfully said to be able to be said to be meaningful efficient core screenings maybe even maximally efficient core screenings of what is happening i think we do have that and i think you can find this even within uh these labet paradigms if you didn't uh i think you still could have many of the varieties of free will worth having because you know many people have criticized it saying like oh just moving your hand without a reason that's not really agency that's not what we want from free will we want like meaningful decisions sure but i i think even though even though for these even for these studies um which people often will you know trot out and say look my brain made me do it even then i i don't think it follows that uh they show that we um that our consciousness is epiphenomenal um through and through and i think active inference basically shows some of the particular ways that consciousness is indeed causal even in these circumstances and i think that's a very big deal uh i i don't know how well it's surviving the replication crisis but you know there's plenty of studies uh showing that uh some undesirable consequences of people uh not believing in their own free will or their own agency uh whether or not that works i think there is a kind of um this face valid account of this and that like kind of like um dumbo's magic feather so like you know this elephant uh and i'm taking this from dan dennett uh from his book freedom evolves but uh the elephant uh has a feather that the crows told him uh we'll help him fly and he had his big ears and he used him to fly when he had the feather and but it was because he believed um whether or not um believing in free will gives you free will not believing you're a meaningful cause that becomes true just by believing it so uh you know i think it's some it's quite dangerous to take basically something where we have something precious something that basically would empower us to pursue our values and to say based on the set of studies and using the authority of science telling people uh that they are not meaningful causes i think that's extremely pernicious um and i think active inference can there's already great work being done like within like the neurophilosophy of free will pushing back against us and i think active inference can help uh showing people what are the meaningful degrees of freedom that they have um and how can they be preserved and expanded wow very interesting do you have any more slides to go through or may i ask some questions uh no more slides uh that was it okay so um those who are watching live please feel free to ask a question otherwise i want pick up near the beginning when you spoke about finding oneself in a stream of action so what is happening when we find ourselves in a stream of and i i want to hear what you have to say and then take this to some questions about memory and free will but when we are suddenly aware like whoa i'm on a run i mean i guess i did make prior decisions about deciding and so on but like whoa now my leg is moving forward so the i imagine this this varies a good amount like as a in in pure and intra individual difference variable so like both between and within people you're going to need a good amount of variance on this i think but the way you know heidegger described it in terms of this difference between like describing like hammering and you know when you're a skilled crafts person uh it's the hammer is in your hand and it is ready at hand and you're you could your mind could be elsewhere or people might have the phenomenology of this like on a run or on a drive you might just find yourself home you can listen to an audio book you'll barely know you're just you have this effortless mastery of the task set and it's only when something goes wrong that things are brought into awareness the it's only and so then you get this motion between so here's the hammer just in your hand ready at hand to present to him and now it's in the foreground it was initially in this um unconscious or semi-conscious background and so if someone's running like for instance they could largely be in a similar state of effortless mastery especially because you know just natural pendulum motion gets you like a lot of the control um that combined with some spinal pattern generators you know it's and a little other tweaks it's like you barely need to do anything once once you've been trained up uh but let's say you're trail running and or let's say you run and then you step on something the idea would be that this violation of your expectations uh would cause uh basically prediction error and that this prediction error the the thing that was different so okay so within predictive processing part of one of this targaryen phenomenology where things just become unconscious and you only become conscious when you're um something's wrong you might while being careful not to conflate levels of analysis you could get a good amount of this just from predictive processing with the dual tier architecture like if basically you're really well trained up you can explain things you can basically handle things just with these efficient unconscious couplings unconscious uh as a in a bunch of reflex arcs being skillfully deployed doing most of the things you need to do and whether you want to call this modeling or not you know be a matter of choice i would call it modeling but you wouldn't need to become aware of unless something was wrong and so then the idea is this lower down it's not handled you don't predict if you don't explain away the prediction error it makes its way deeper into the brain into it becomes more broadly the deeper portions of course explicit modeling and conscious modeling processing challenge prediction errors will be greater more capable of accessing these richly connected sub networks which could support basically this larger scale integrative simulation some aspect of which would be your i don't know if that got it at all but yeah awesome so we're we're driving back towards free will because i really liked how you approached it in in this conversation so i want to think about three concepts memory free will and experience what are their venn diagrams like can we of things that we remember did we have free will for them or not more or less than expected for things that we experience do we remember them more or not could there be something that you didn't experience but you remember and had free will on and so on like how our memory free will and experience bound together and that's a fantastic question i don't so what's coming to mind that's that that's a whole other conversation though what's coming to mind for me at the moment is um there's a partial definitional aspect of a connection between free will and experience and that what people so people want different things i wouldn't call free will one thing um in fact i actually think there's a certain wisdom and just like the language of it like people want both like the the willing they want the agency and they want the degrees of freedom in terms of they want to be able to pivot they want to be able to have what they do be a function of their past and being constrained by their memories and their meanings from the past but they also want to be able to break free um whether in the face of something new or just encountering something new or to explore and so um having that that freedom for exploration is also going to be a part of agency both within a context and across time but the memory is going to be important because without the memory uh that's the constraint based on values based on you um you know there's the many aspects of the cell for instance uh some some of the most essential ones and maybe some of the the gold standards of free will some frankfurt mesh theory afraid it was but it's like this interface of like you both want things and you want that you want them where the wanting that you want is based on this integrative and like narrative type self and so that would be a way in which memory would very strongly like where free will be a function both of experience that you want your experience your consciousness and your experience to be part of the causal stream leading to action part of your your valuing your affective inference you want that to be influencing both across time and in the moment um and so the experience would be essential like it's not clear that many like dennis has argued like that um the freewill consciousness must always be handled together um but and without without memory um you also though it's not you either i think there's an interesting thing though um in terms of like you're alluding to like basically connections between causation and memory uh and i believe we are better capable of remembering things when we can take them up into a causal unfolding this gives this this basically affords a representation with a more of a sparse structure of of salience of this led to this led to this and then you hold that causal graph in your mind and as basically the core of your generative model and it seems to be like a highly efficient way of remembering things is come up with a causal account of them and so if something is a product of your agency in a way that's that is all automatically structuring things in this causal set fashion and so there should be a connection between having agency and free will in a circumstance and being better able to remember things i don't know if that's the case but um it seems like some things from education theory suggests like with respect to active learning but i could see that going a lot of different ways um but and so that's a deep question and those that's that's what came to mind but like what do you think wow well the the last piece you brought up about the the sparsity kind of the sharpness on the sparsity of the causal story and then things that have a sharp and a sparse causal architecture as experienced with our very limited working memory they're like pre-digested and therefore very apt to be able to leave certain kinds of traces and then maybe something could leave a trace that didn't have a perceived sparse causal architecture in the moment but that might be like an example of a memory that's hard to recall or that might be like cleaned up in post-production and maybe very susceptible to being influenced because like subsequent updates could um try to draw attention to part of that uncertainty and um like if there was just a loose memory of many things happening at a scene asking even a question about part of it could start to sharpen that aspect so it's it's an interesting question and also kind of like the questions like what if you had some sort of suffering but you didn't remember it or in between moments or when you were sleeping you were having some sort of negative so just i mean what is happening with the mind and the brain the the brain seems continuous but then we have these more flashes and discreteness and shifting of mental attention it's good i just like the way you're describing i feel like it's like like a really good case in point of like the answer to do we or don't we have free will should always be what do you mean and and the answer is always like um like you like the way like you're showing these different processes could either go together synergistically or come apart um how many meaningful degrees of freedom do we have and what circumstances or and what senses do we have how much of which of the degree varieties of free will that we want in different circumstances so it's like the answer is there isn't their free will it seems like like much more interesting is getting into the details and useful it's like getting into the details you're talking about right now with like the different ways that memory and experience and choice can all go together or not both within and across time like that's and then that answer is like that i guess you could say that's like that's the real problem of free will and then that that um interesting yeah like um that when the experiment of pushing the button is presented as being about the exertion of decision making in the moment rather than the um memory or the recalling of something that might have been different in the moments those are quite different settings and it could be a total different neurophysiology so and also what you said about how this sort of tension between having a connection to the past and the constraints that it provides but also the ability to choose and i guess that's what's explored in these varieties papers so i wanted to uh not just ask like what are the varieties but just specifically asking about the similarities and differences with bodily action what are the varieties of free will we might have with respect to bodily action like the so-called voluntary and involuntary bodily processes and then mental action where in active inference we can model regimes of attention and metacognition and thought as like action policies on mental states so what kinds of free will are out there for the body does that say about what might be a cognitive degree of freedom or not like you know just choosing what we want in terms of like body actions if we're dealing with something highly rehearsed then it seems like that could be an example or or something where it's like you can model that the predictive processing like a hammer strike and the loop is closed before it ever even enters like the brain it's in the um cn the cns and the reflex arc but doesn't perhaps even make it up top where your body moves that's just after the fact that you're aware of it but then if you're doing something like uh and so then like at this other extreme of abstraction you might have something like planning uh reaching a distal goal where there's many steps involved and you need to come to some sort of sequence of the steps and along the way um actually i don't that'd be the extreme but that would be a little like there's certain things that have to happen in some ways like at the extreme of abstraction it might be something like uh going to grad school or something like that or you know uh too real winning a new job two ouch um but like that itself there's a lot of things that happen have happened before other things a lot of dependencies and a lot of multiple realizability like there's always multiple realizability like you know i go to reach for this glass i could have reached for like this i could have done a weird thing and gone like this or like you know i could have pinky out pinky n you know i prefer pinky out um but like there's all sorts of things that could have done there is multiple realizable but the multiple realizability of something like you know going to college going to grad school that's even more so like the number of ways you can go into it and then i guess i'm thinking like in this middle regime of something that's still like very clearly metoric but would be novel in that like um fine-grained motor control like i'm learning for the first time like i'm doing a tea ceremony or i'm learning a calligraphy or i'm just learning to knit or something like there it's the fine-grained things are now being monitored um as uh present at hand and and you actually need to use your sophisticated inference machinery to help scaffold this basically um the acquisition of skilled motor abilities of skill the capacity for skilled motor performance i would also i think suggest that um the whole way through it's always built um experience when when once it's being co-opted or controlled in its controllability is always um modal in a motoric fashion uh so one of the uh claims i make in this paper it um out that uh it's called pre-motor theory but that basically like uh through partial expression of a motor command or a motor prediction the reference copy from that the associated sensations can be used as a means of top-down attentional selection could be used as a means of attending to memory in terms of working memory basically you can use these partially expressed motor predictions not enough to actually cross threshold but use them to basically as a source of control over dynamics um of where your awareness is going and what's going in and out of your awareness and the claim i make is actually that this is uh exhaustive like this is the only like lever you have is basically whatever is built off of a control hierarchy off of skeletal muscle it's a claim the idea would be like even talk about something as abstract as you know going to college or grad school there's still going to be some so it has to cash out as attic and modal you're seeing yourself in the different contexts even for the abstracts like the most like justice um you you still to experience it it's still going to have some sort of thing that you you know here's justice's scales here's the courtroom here's seeing this memory from your past where there is an impulse or justice was served here's you feeling this like uneasiness or this like righteousness um so the idea i would claim is that like any kind of control on any level will always cash out motorically but there are meaningful differences in terms of um whether you're engaging with something which is involving these immediate fast feedback loops it might be multiply realizable but less so or something which the um affordance structure is more being played out in terms of your capacity to control it and relate to it in terms of these even more multiply realizable high-level plans of all the things situations you might be in they'll be related to achieving high level uh goals or abstract goals i don't know if that makes sense well one one thought on that is it's like a jigsaw puzzle so there's only one way to assemble it and it's kind of like there there's therefore this bundle of trajectories of different realizations of how one is going to assemble it like there's sort of what happens on the table all the combinatorics of the way to assemble it and then there's like all of the real-life combinatorics about how many minutes am i gonna spend playing this jigsaw and then i'm gonna go get up and do other stuff and so the multiple realization of the jigsaw where it does all kind of come back to being completed is one thing and then yeah you brought up how when things are even the outcomes are unknown or there's multiple realizations those are where we start to maybe think about hierarchical nesting of models like we've been discussing in the slam paper and also the vivid imagining of different future states what is the regime of attention that might for somebody who wants to be an athlete help them make practicing and being on that path like more natural so what what how specific and what modalities that's kind of what it made me think of but i i kind of want to keep on going on this mental action route has there been anything like a mental or a cognitive libay experiment like where the instructions would be like take a couple of breaths and just you know get up and walk around or whatever just take a couple breaths and then identify the moment on the clock when you chose to think about x a pure mental on mental assessment just before i forget i actually think like the what you're just mentioning like the optimal intentional engagement for something like motor or something like like a sport or some performance actually that's like a key example of through skilled mental action or not through the skillfulness of your mental actions having more or less freedom where like for instance um if you're choking for instance you do it wrong um june tony has a good paper on choking and active inference and so if you get that wrong that would be like your free will being undermined by your mental action not being uh well calibrated in terms of where you're positioning your intentional selection for optimal grip or optimal engagement with what you're doing in terms of this pure like more purely uh mental action the bed and the there's a couple there's different um versions of it that have been run uh some of which i think would get a little bit more to that um kind of what you described in terms of um uh uri mouse has one paper called um decisions like it's deciding or decisions that matter and so basically it would be choosing between charities and so then you so and then in this charity choice you actually don't see this same type of ramping and i think the support to some degree um the model i'm suggesting and that like if this ramping is a buildup of basically activity in your action-oriented body maps cross-coupling with your interceptive body maps i mean this feeling of urge and mental stimulation it makes sense you would have that for something like pressing a button but if it's something like which charity i'm going to donate to um you might not get that same kind of tight coupling and feedback and so i think that's some evidence in support of that but i think the particular thing you brought up there's these um this is a direction i'm uh looking in beginning to look into there's these closely related waveforms to the readiness potential um the um event-related negativity and the um spn um stimulus preceding negativity and i believe so some of them would be for instance um preparing for information that's coming um and then acting i think that would be the ern or the spn the stimulus proceedings you're preparing to receive information that's possibly salient but there's no action required so i think that might be potentially you could think of all these things though inactively as having they look very similar electrophysiologically and i would suggest that they are all fundamentally similar and that what they're using whether we're talking about um patterns of attentional selection or specifically um orchestrating mental stimulation to act or not in terms of an overt action in all these cases it is grounded in the modalities and this is basically understood implementationally as recurrent activity in different places of control hierarchy over body maps or algorithmically and functionally an accumulation of model evidence for shifting your proprioceptive pose now before an overt action but for talking mental action similar story but if partially ex but if if the model evidence i guess would be a slightly different model but it's i if you if what you're imagining is your own you're imagining your body being in a different state that would allow you to orient differently to bias your sensors differently to move your attention around differently to do different precision weighting you you don't imagine it vividly enough that it will actually drive your spinal reflex arcs and that but that this would be what you would use even for potential purely mental acts it would still be co-opting a control hierarchy off of skeletal muscle and the reason i focus on skeletal muscle is because you know it's designed to have basically large gross actions with low latency and it's controllable and and and the conditions for fine-grained control are realized because you can actually direct your sensors using your effectors um with a large-scale meaningful outcome that can be causally modeled and tracked and controlled and so then the idea is that basically vygotsky first expressed an idea like this i think with respect to thought is inner speech where first you learn how to speak and then through partially expressing your speech you learn how to co-op this process for a stream of thought um as inner speech and then whether you want to say that that exhaust thinking is a whole other thing but inner speech would be part of a significant part of thinking um but so the idea is this is all intentional mental operations whether we're talking about um shifting around basically the the fine green adjustment of an overt skilled motor act or guiding your mental action with respect to attention and mental imagery and imaginings that in each case you have the initial affordance structure the initial control levers established via these overtly expressed behaviors and couplings of the world which then later by partially expressing them gives you something you can repurpose uh for things like imaginative planning or covert detention it's or i would argue any intentionally influenced mental process may be ultimately have its origins in this co-opting of motor control bold claim hmm so what systems yes go ahead i just said could be wrong not financial advice not not extraterrestrial advice hardly even active advice though it's very interesting um what systems rise to this capacity and does their embodiment matter so like there are insects with tremendously reduced central nervous systems um one could imagine a uh computational system that can engage in counter factuals but it's only affordance or embodiment is like one you know morse code receiver or something like that so what have this capacity and why um so i i definitely don't know the boundaries but if i were to shoot from the hip on this the idea would be the handling would still be of this computational neural phenomenology or mario phenomenology but instead of drawing brains uh we would draw a different nervous system or whatever the um signaling information processing system is we would still give it this kind of handling where there would be a mechanism there would be a function there would be some intermediate algorithmic mapping across these levels and then you connect this to phenomenology and so if in this phenomenology you're having basically goal states being represented and these and the representation of these goal states um is part at that level you're also able to get an account of the guidance of action selection to bring them about then you could say that you have degrees of freedom in that way so for talking about something like in insects um they uh you actually sent me a really interesting paper recently about dual systems thinking broadly construed um including for insects where and this it seems like you would need at least a a two-tier hierarchy where this upper level can break free of immediate engagements with the censorium and where this lower level would allow for your couplings with the world and the orchestration of overt action selection uh in terms of actually like sensory motor cup into the world and then a process which can basically top of that hierarchically higher that could infer these state transitions but but without them being expressed and if you have that then you could do the type of counter factual processing that you would need for planning and constructing causal models it seems possible that insects might be able to do this in terms of um for instance like in that paper you sent uh recently um these like do this so the central complex uh has like a lot of these properties of the hippocampal system um it's kind of like it has properties of the hippocampal system and the thalamus at the same time and it's giving you like the salience map used for both orienting and moving uh but that you also get these mushroom bodies which are like the stratum which would be basically helping with moment-to-moment action selection biasing it in a value driven direction uh in this paper you're getting like dual dual systems for control one which seems to be more immediate and reactive and the other seems to be governed by slower processes and so it's possible that something like this so it's possible that this could give you the ability to do sweeps of possibilities of counter factual inference where you basically take the beginning of expressing the motor command so the insect is thinking i'm a locomotive in that direction um i i don't know the evidence well enough to know if this is the case but it could be so so for a like a rodent you'll see these sweeps in the hippocampus of like there's a maze and you'll see like predictive of whether the mouse goes left or right you'll see these place fields representing different points in space and in anticipation of this you'll see these sweeps ahead of activity and then where this sweep is most robust that'll be the branch that like the animal will go down we'll go down back to the places in theory you could do a similar thing by other means uh maybe with some homology even in the insect nervous maybe you can even do it in an ant colony if you're getting like an inner attractor capable of aggregating across all these different finer grain couplings and engaging in control but so basically whatever has uh or what the i i can't say i'm not going to say sufficient but the claim i'll make for necessity would be some sort of dual-tier architecture where the lower gives you the couplings of the world and the upper level gives you counter factual affordance and integration uh across uh these lower level potentially segregated um coupling processes something like that could could do it for you and so could insects do it oh sure potentially i don't know could the uh insect colony do it maybe the thing where i would place a potential constraint it seems like we might need an to have properties where it could signal fast enough where these inferences of quickly enough that it can stay on top of its evolving action perception cycles so in terms of its surfing uncertainty it's moving through the world as couplings the architecture whether we're talking about a brain or some sort of signaling across multiple individuals like a hive mind we wanted i think to be fast enough have the estimates correspond to this sparse graph um what benjamin would call i think independently controllable factors of the world a causal world model where you are an agent within it but it seems like you would need to have the ability to model um independently controllable factors and so you need a signal signaling system that's fast enough to pull this off and so some sort of small world network architecture would be likely required where you have a combination of local modularity but then synergy through having some central hub and so if this could be pulled off by the hive mind and if this can orchestrate the overall swarm to do things quickly enough to basically have purchase have the right grips on the world that it can both inform and be informed by its action perception cycles then you could have an agency which by some dimensions we might describe in terms of free will that dodged the question before i respond what about just the computational one what if it was a computational one and it was it was instantiating a digital twin a limited um or comprehensive digital twin of itself i mean i personally think um the the i feel like like an integrated information theorist would say absolutely not some people uh within like active inference communities would say uh no it has to be physically realized but i think something like a digital twin or some sort of emulation um the important thing for me is that the there is a set of dependencies of effective connectivity that could be involving virtual processes and that if you trace out the patterns of effective connectivity of kinds not necessarily physical necessarily like made of matter if that is entailing the modeling of a system and its relationships with the world and where within that system world model it has this kind of agentic affordance uh for me that counts but a lot of people would disagree on that but for me i think you're trying to find the maximally explanatory core screening and sometimes that might be fine found at the level of the virtual machine and not in terms of the physical realization and then that's actually the most meaningful cause and then if you trace it out there that's where you would look for it and so that could be in a digital twin or uh emulation but a lot of people would disagree with that so when you were talking about applying this to the colony level i thought about like the foraging behavior that's like reaching out of the ground and grabbing a seed and so that's like the enacted foraging trip is there a counterfactual i could be foraging or we could be foraging at some different scale underground uh on the other side of the markov blanket of the nest entrance and then what if somebody were going to be tracking the the elevation of a forager and they'd be like well they go up and down in the nest but on the times where they make that decision to forage they always go up 200 milliseconds before foraging it's like well yeah that's where the nest entrance is so without going into whether they have will or not it's some kind of cognitive motor integration when the person hits the button so then the way that you frame this as being like not a trivial artifact in fact an interesting outcome of aggregation of erps across trials how couldn't there be a coherent preamble now i guess it doesn't if we're 30 seconds long that would be a different setting but then when we connected to the specific mechanisms by which they're that do have to come together in order for there to be a coherent motor action it's it's quite an interesting view and i wonder how those who have um interpreted it in a very different way feel have you spoken or asked to anyone who has a very clear and different interpretation of what this potential means i've been discussing it some with um before i forget i've been discussing some with them or a good amount with um erimos and aaron sugar aaron sugar has he has multiple views but some of them do disagree and so one of them is you in theory because you're averaging across multiple trials there is an interpretation of readiness potentials as artifactual um and it's possible that sometimes you are dealing with an artifact of measurement and this is like an open question but he's also uh for instance uh uh very like he's he's testing out like the the mental simulation view also um another interesting hypothesis he has is actually some of the evidence accumulation is actually fast feedback loops with muscle spindles and actually you're getting basically the the the urge actually a lot of is actually in the body itself uh but one interesting thing before i forget is that these um build up and then threshold crossing i believe this you'll see across all nervous systems where you get action selection and i would wonder whether this you know similarly similarly to me saying like you know this isn't just like a brain story might we see basically something like this if we're looking at them i think you're active in paper you talk about like stigmatic coherence but through some sort of like forgive me if i'm like butchering it but like some sort of like measure of that kind you might see something like generalized action potentials for large scale changes in switching modes of the hive mind of the colony uh i and while i'm just speculating i actually also in terms of multi-stage hierarchies um i actually wonder whether um the brains like for instance the resting state networks of the brain maybe not like there's different like granularities and parcellations you can be as much as 17 or or more if you wanted but um there are some solutions that are like maybe you know so seven networks are most well known but there's also like five network solutions and i'm wondering if they're actually universality classes for active inferential agents that you'll actually get something like homolog of different western state networks in their patterns of effective connectivity so generalized wrestling state networks generalized empirical phenomena to like things like readiness potentials uh very uh very speculative but that's uh that's the i think we might a few interesting things that makes me think of and uh are we making cognitive science a creation in our own image if we talk about the generalized hippocampus and the general and so to what extent are we or can we must we generalize from our own action perception loop is uh putting another umbrella which the human mind is just a realization of and saying well the cognitive domain is bigger than the human brain because it includes the brain body and the extended niche but also includes non-human systems like if we want a cognitive science that includes a thermometer or something like that like a thermostat is that a is that a cope because actually that is still the water that that model swims in is still human and cognitive i don't know but these are interesting pieces i mean it seems like it seems like surely what you're describing has to be the case and it's and that's like both a good thing and a bad thing where it's like we need to be able to draw on an analogy with something we understand like we're always like we're usually making you know sense of things in terms of other things and so like your our own experience as this base domain in order to give us some purchase like can we do it without it you could say like first principles thinking but like which first principles um but then there's like all of the like the baggage that could come like the disallowed like if you're if it seems like we can't avoid doing that maybe we could but if we want to but then like we really need to be careful to track the dis analogies if we're saying generalized hippocampus like okay wait wait wait what are the dis analogies here what are the proper unique properties that are present in the target domain and not the base domain of like the map the analogical mapping you're doing or vice versa yes yeah that's fraught what you're describing well i think a kind of closing area just one point that i want to pick up on i thought it was really powerful what you said about really the social consequences and the personal consequences of interpretation of neurosciences and in some ways even science communication around empirical results the idea that we could opt out to free and that in a world where there's limited degrees of freedom you might be coerced into thinking you had free will or not but if there's any glimmer of free will it can be used to deny and deceive itself so that puts research in this area a different reflexive position than research on other systems so i guess just how do we move forward here while drawing on the specifics and being respectful of basically the past the present and the future i mean for me i definitely have a lot of uh soul soul in the game and that like this is actually maybe the biggest thing a couple things but this might have been one of the biggest things that got me into cognitive science was a very long extended um freak out about free will when i was younger where it was just i couldn't see how anything could be mean i couldn't see how anything could be meaningful without it um i don't know if i still believe that exactly but for me it was it was very difficult to not believe that there were like this sort of all or nothing thinking about free will and then going with the nothing dennett actually rescued me from that with freedom evolves that his book but uh there was a long time where it seriously bothered me and it had real negative consequences in my life the and i'm not alone in this i've talked to others and they also feel that way and um part of the way this is coming up in the present work with um psychedelics actually is um so there's this study with um i'm preparing for with creative professionals who are suffering from burnout one thing i'm wondering in terms of like people can get burnt out from you know multiple reasons to the extent that this is continuous with burnout is continuous with depression or like feeling like you don't have agency in context where you want to have agency um or or feeling like you don't have that you know adequate control for something where it really matters that you do that seems like a good recipe for burnout and it seems also like a decent model for some forms of depression where uh if you view yourself as not being able to be a cause in the world if you're not able to meaningfully work towards your values that could be very distressing i mean it could also be a state of like a quasi-buddhistic like blissful surrender you know there's a question are we flirting with nihilism there this question uh and the other question is uh you know is it even legitimate though to be an eliminativist on the self and on volition and then just to say it's okay you're not real your vision isn't real anyways don't worry about it was that even a legitimate intellectual move to begin with i would argue it wasn't so yeah i think this is the stakes here for people sense of themselves i believe they really matter and i believe they have both social and clinical and social consequences existential consequences for people and i'm glad that uh there's this like growing community of people in the neural philosophy of free will who have been doing a really great job pushing against like some of these i believe excessively inflationary views and i believe that active inference might be a really valuable contribution to these efforts in terms of providing formal models and models that can go across this marian neurophenomenological stack um or computational phenomenological setup and actually address these phenomena in enough detail where not only do we feel meaningfully satisfied where we understand our own agency that we could we could intervene um and hopefully for the better yeah just just some last thoughts on that that's very powerful and the the mutual constraints and the interdependencies of the different levels like it may come down to someone's genetic sequence of a receptor or um some kind binding affinity with a molecule but then it could also be integrated in kind of a full stack way talk or action or changes to the niche so it gives like a full stack yes and way that's not uh i think a dead end it's more like a starting state like if the computer weren't working and then there was like the hardware and there's the software but then moving beyond just that type of analogy just very powerful about burnout and about the different cognitive ways that we might get there and get into it and get out of it and so it's like stoicism or whatever plus preferences because it's not just the neutral stance it's not just a descriptive stance on burnout there's like a preference there so adam if you have any other thoughts please feel free i guess a quick thought is if we're looking and this is like to follow up on the um commentary saying like these somewhat buddhistic eliminativist framings of um agency and i said like are they even legitimate uh i i'm gonna argue um no they're not uh and specifically um if we're thinking of something like modeling the self through time as a high level controller if you um engage in some sort of practice whether it's discipline meditation for a long period of times or you've taken like a very high dose of psychedelics and you've basically collapsed the depth of your the temporal depth and counterfactual richness of your generative modeling you will not find yourself there you will be gone there will be no like you to talk about and you might call this something like hugo death but then to say that there never was a you process to begin with um i believe that is actually illegitimate you've basically used a search strategy um that a priori should not have been able to find the self and then using that search strategy you then concluded it didn't exist and um i think that just doesn't work um in this case you know the quote down and again uh you know individuals their narratives their persons you know they're real patterns and um but they're also there's a reflexiveness to it where their own belief and their own reality is an important part of their causal significance um you know i think uh it would be good to have some some nice debates with like different members of the active inference community who are more deflationary on things like selves self processes and agency because ultimately i think this is a thing where it's going to be like well we should you know be respectful and disagreements um i think it really matters we get it right to the extent we can thank you adam for the excellent presentation apologies for any internet glitches that might have happened we are a strange loop and this has been a strange live stream always good talking to you daniel